

Jadavpur University

Department of Information Technology



SOFTWARE LAB – II · LABORATORY REPORT

Design and Implementation of a Project Management System

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1. Problem Statement

Design a system, using object-oriented principles, for a company that handles multiple clients (projects) and assigns a specific set of employees to manage particular projects. Implement the whole system and generate a UML representation of it. The system must provide the following features:

1. assignment of employees to projects, recording the employee, the designation and the skill;
2. the project requirement, raised by a general department;
3. a project development tracker covering the life cycle of requirement and feasibility, testing, deployment and maintenance;
4. project billing; and
5. client feedback at each stage.

`Project` is to be taken as the super class, and further sub-classes are to be created as required.

2. Objective

The objectives of this assignment are:

- to identify the classes of the given problem domain along with their attributes, methods and mutual relationships;
- to represent the design as a UML class diagram;
- to implement the design in Java, using `Project` as an abstract super class;
- to validate the implementation with a set of test cases; and
- to build a graphical front end that operates on the same model.

3. Theory

The design applies the four standard relationships of a UML class diagram. *Inheritance* is used where one class is a specialised form of another. It allows the abstract operation `complexityFactor()` declared in `Project` to be redefined by each concrete project type, so that the billing module obtains the correct multiplier without knowing the actual type of the object it holds. This is *polymorphism*.

Aggregation is a whole-part relationship in which the part can exist on its own. An employee continues to exist if the company record is removed, so the link from `Company` to `Employee` is an aggregation. *Composition* is the stronger form, in which the part cannot exist without the whole. A `Billing` record has no meaning apart from the project it belongs to, so that link is a composition.

An *association class* is used when the link between two classes carries information of its own. The statement "which employee handles which project" is not merely a link, because it also carries the role the employee plays and the date the allocation was made. It is therefore modelled as the class `Assignment` rather than a plain association.

4. System Design

4.1 Identification of classes

The company is taken as the root of the system. It holds the clients, the employees and the projects, and it performs the allocation of an employee to a project. Each of the five features listed in the problem statement is given a class of its own, so that no single class carries unrelated responsibilities. The classes identified are listed in Table 1.

Class	Attributes (principal)	Responsibility
Company	name	Root class. Holds clients, employees and projects; allocates employees to projects.
Client	clientId, name, contactEmail	A customer of the company. Owns one or more projects.
Employee	employeeId, name, designation	A member of staff holding a designation and a set of skills.
Skill	name, level	A named competency held at a stated proficiency level.
Project (abstract)	projectId, name, status	Super class. Holds the requirement, tracker, billing, assignments and feedback of a project.
WebApplicationProject MobileAppProject DataAnalyticsProject	—	Concrete project types. Each supplies its own complexity factor.
Assignment	roleOnProject, assignedOn	Association class joining an employee to a project.
ProjectRequirement	department, summary, feasible	The requirement of a project, owned by a general department.
DevelopmentTracker	currentPhase, phaseCompletion	Tracks the current life-cycle phase and the completion of each phase.
Billing	baseCost, loggedHours, taxRate	Computes the invoice for a project.
ClientFeedback	phase, rating, comments	A rating and comment attached to one life-cycle phase.

Table 1: Classes identified in the problem domain.

Four enumerations support these classes: `Designation`, `LifeCyclePhase`, `ProjectStatus` and `SkillLevel`. Keeping the life-cycle phases in an enumeration fixes their order, which the tracker relies upon when it advances from one phase to the next.

4.2 Relationships between classes

- **Inheritance.** `WebApplicationProject`, `MobileAppProject` and `DataAnalyticsProject` extend the abstract class `Project`.

- **Aggregation.** Company aggregates Client, Employee and Project, each with multiplicity 1 to many.
- **Composition.** Each Project is composed of one ProjectRequirement, one DevelopmentTracker, one Billing and many ClientFeedback objects.
- **Association class.** Assignment joins Employee and Project with multiplicity many to many.
- **Dependency.** Classes depend on the enumerations that type their attributes.

4.3 UML class diagram

The complete class diagram is given in Figure 1 on the following page. As the diagram is wide, it has been printed sideways; the page should be turned clockwise to read it. It was prepared in PlantUML, and the source file `uml_class_diagram.puml` is included in the repository.

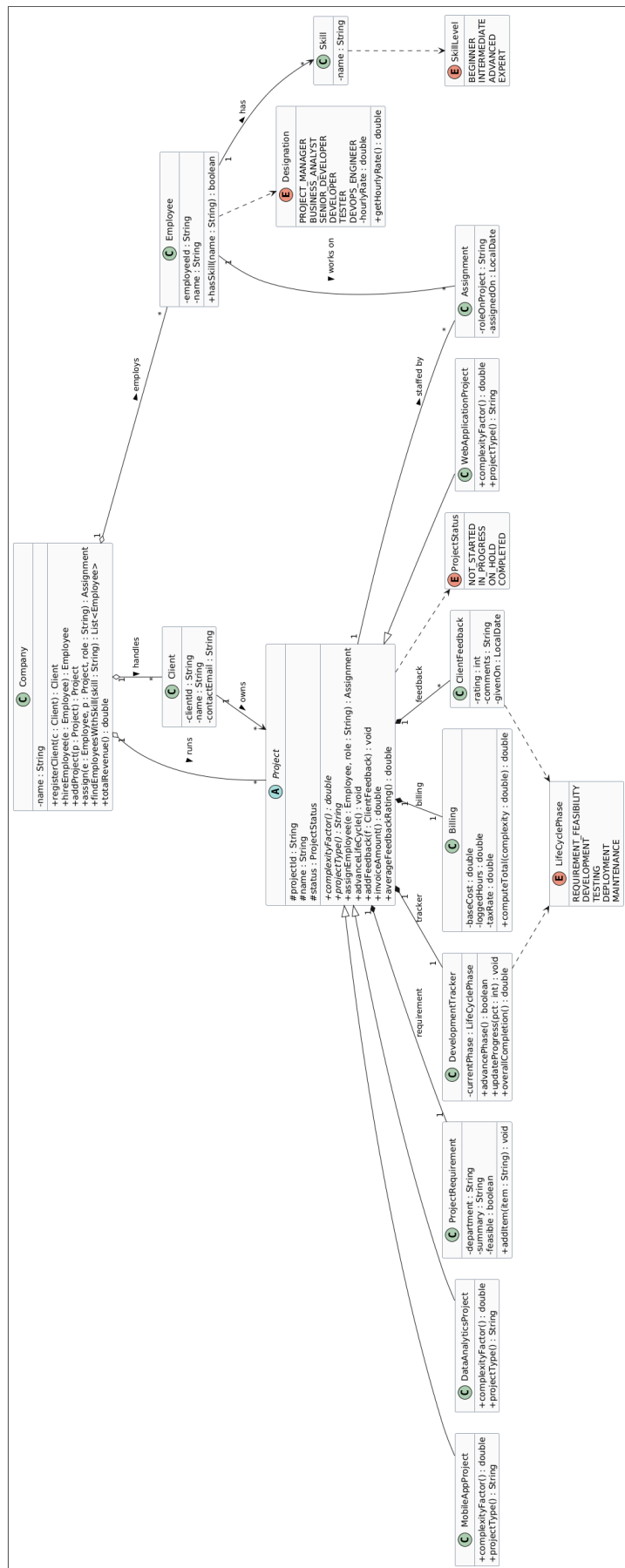


Figure 1: UML class diagram of the project management system.

5. Implementation

The system is implemented in Java (JDK 11). The source is divided into three packages. `com.pms.enums` contains the four enumerations, `com.pms.model` contains the domain classes, and `com.pms.app` contains the demonstration driver and the test suite. A fourth package, `com.pms.ui`, contains the graphical front end described in Section 7.

5.1 The super class

`Project` is declared abstract. It owns the requirement, the tracker, the billing record, the list of assignments and the feedback log, and declares two abstract operations that every concrete project type must supply. The method `invoiceAmount()` calls `complexityFactor()` without knowing which subclass is executing, which is where the polymorphism of the design is exercised.

```
public abstract class Project {
    protected final String projectId;
    protected final String name;
    protected final Client client;
    protected ProjectStatus status = ProjectStatus.NOT_STARTED;

    protected ProjectRequirement requirement;
    protected final DevelopmentTracker tracker = new DevelopmentTracker();
    protected Billing billing;

    protected final List<Assignment> assignments = new ArrayList<>();
    protected final List<ClientFeedback> feedbackLog = new ArrayList<>();

    protected Project(String projectId, String name, Client client) {
        this.projectId = projectId;
        this.name = name;
        this.client = client;
        client.addProject(this); // keep the association consistent
    }

    public abstract double complexityFactor();

    public abstract String projectType();

    public void setRequirement(ProjectRequirement requirement) {
        this.requirement = requirement;
    }

    public void setBilling(Billing billing) {
        this.billing = billing;
    }

    public Assignment assignEmployee(Employee employee, String roleOnProject) {
        Assignment assignment = new Assignment(employee, this, roleOnProject);
        assignments.add(assignment);
        if (status == ProjectStatus.NOT_STARTED) {
            status = ProjectStatus.IN_PROGRESS;
        }
        return assignment;
    }

    public void addFeedback(ClientFeedback feedback) {
        feedbackLog.add(feedback);
    }

    public void advanceLifeCycle() {
        boolean advanced = tracker.advancePhase();
        if (!advanced || tracker.getCurrentPhase() == LifeCyclePhase.MAINTENANCE) {
```

```

        status = ProjectStatus.COMPLETED;
    }
}

public double invoiceAmount() {
    if (billing == null) {
        return 0.0;
    }
    return billing.computeTotal(complexityFactor());
}

public double averageFeedbackRating() {
    return feedbackLog.stream().mapToInt(ClientFeedback::getRating)
        .average().orElse(0.0);
}
}

```

Listing 1: Project.java, the abstract super class. Accessor methods and toString() are omitted.

The three concrete project types differ only in the complexity factor they return and the name they report.

```

public class WebApplicationProject extends Project {
    public WebApplicationProject(String projectId, String name, Client client) {
        super(projectId, name, client);
    }

    @Override
    public double complexityFactor() {
        return 1.2;
    }

    @Override
    public String projectType() {
        return "Web Application";
    }
}

```

Listing 2: WebApplicationProject.java. The other two subclasses are identical in form, returning 1.4 and 1.6 respectively.

5.2 The association class

Creating an Assignment registers it on the employee inside the constructor, and Project.assignEmployee() adds it to the project. Both ends of the association are therefore always consistent, and the model cannot reach a state in which one side knows of a link that the other does not.

```

public class Assignment {
    private final Employee employee;
    private final Project project;
    private final String roleOnProject;
    private final LocalDate assignedOn;

    public Assignment(Employee employee, Project project, String roleOnProject) {
        this.employee = employee;
        this.project = project;
        this.roleOnProject = roleOnProject;
        this.assignedOn = LocalDate.now();
        // keep both sides of the association consistent
        employee.addAssignment(this);
    }
}

```

Listing 3: Assignment.java, the association class between Employee and Project. Accessors omitted.

5.3 The development tracker

The tracker holds a completion percentage for each phase and a reference to the phase currently in progress. Because `LifeCyclePhase` is an enumeration, the next phase is obtained from the ordinal of the current one, which guarantees that the phases cannot be visited out of order.

```
public class DevelopmentTracker {
    private final Map<LifeCyclePhase, Integer> phaseCompletion = new LinkedHashMap<>();
    private LifeCyclePhase currentPhase = LifeCyclePhase.REQUIREMENT_FEASIBILITY;

    public DevelopmentTracker() {
        for (LifeCyclePhase phase : LifeCyclePhase.values()) {
            phaseCompletion.put(phase, 0);
        }
    }

    public void updateProgress(int percent) {
        if (percent < 0 || percent > 100) {
            throw new IllegalArgumentException("Percent must be 0..100");
        }
        phaseCompletion.put(currentPhase, percent);
    }

    public boolean advancePhase() {
        phaseCompletion.put(currentPhase, 100);
        LifeCyclePhase[] all = LifeCyclePhase.values();
        int next = currentPhase.ordinal() + 1;
        if (next < all.length) {
            currentPhase = all[next];
            return true;
        }
        return false; // already at MAINTENANCE
    }

    public double overallCompletion() {
        return phaseCompletion.values().stream()
            .mapToInt(Integer::intValue).average().orElse(0);
    }
}
```

Listing 4: DevelopmentTracker.java. Accessors omitted.

5.4 Billing

The invoice is the sum of a fixed base cost and the labour cost, multiplied by the complexity factor of the project and increased by the rate of tax.

```
public class Billing {
    private final double baseCost;
    private double loggedHours;
    private double blendedHourlyRate;
    private final double taxRate; // e.g. 0.18 for 18%
    private boolean paid = false;

    public Billing(double baseCost, double blendedHourlyRate, double taxRate) {
        this.baseCost = baseCost;
        this.blendedHourlyRate = blendedHourlyRate;
        this.taxRate = taxRate;
    }

    public void logHours(double hours) {
        this.loggedHours += hours;
    }

    public void markPaid() {
```

```

        this.paid = true;
    }

    public boolean isPaid() {
        return paid;
    }

    public double computeSubtotal(double complexityFactor) {
        return (baseCost + loggedHours * blendedHourlyRate) * complexityFactor;
    }

    public double computeTotal(double complexityFactor) {
        double subtotal = computeSubtotal(complexityFactor);
        return subtotal + subtotal * taxRate;
    }
}

```

Listing 5: Billing.java. Accessors omitted.

5.5 Allocation of employees

Company refuses to allocate an employee who is not on its rolls or to a project it does not run. It also provides a skill search, which is used when a project has to be staffed for a particular requirement.

```

public class Company {
    private final String name;
    private final List<Client> clients = new ArrayList<>();
    private final List<Employee> employees = new ArrayList<>();
    private final List<Project> projects = new ArrayList<>();

    public Company(String name) {
        this.name = name;
    }

    public Client registerClient(Client client) {
        clients.add(client);
        return client;
    }

    public Employee hireEmployee(Employee employee) {
        employees.add(employee);
        return employee;
    }

    public <T extends Project> T addProject(T project) {
        projects.add(project);
        return project;
    }

    public Assignment assign(Employee employee, Project project, String role) {
        if (!employees.contains(employee)) {
            throw new IllegalStateException("Employee not part of company: " +
employee.getName());
        }
        if (!projects.contains(project)) {
            throw new IllegalStateException("Unknown project: " + project.getName());
        }
        return project.assignEmployee(employee, role);
    }

    public List<Employee> findEmployeesWithSkill(String skillName) {
        List<Employee> matches = new ArrayList<>();
        for (Employee e : employees) {
            if (e.hasSkill(skillName)) {
                matches.add(e);
            }
        }
    }
}

```

```

        return matches;
    }

    public double totalRevenue() {
        return projects.stream().mapToDouble(Project::invoiceAmount).sum();
    }
}

```

Listing 6: Company.java. Accessors omitted.

6. Testing

A test suite was written that requires no external framework, so that the program can be compiled and run with `javac` and `java` alone. It first builds a company of two clients, five employees and two projects and prints a report of that scenario, and then executes fifteen test cases. The cases and their results are given in Table 2. All fifteen cases pass.

No.	Test case	Expected	Observed	Result
1	hasSkill() returns true for a skill the employee holds	true	true	Pass
2	hasSkill() returns false for a skill not held	false	false	Pass
3	Assignment is recorded on the project	assignment present	assignment present	Pass
4	Assignment is recorded on the employee	assignment present	assignment present	Pass
5	A newly created project has status NOT_STARTED	NOT_STARTED	NOT_STARTED	Pass
6	Status becomes IN_PROGRESS after the first assignment	IN_PROGRESS	IN_PROGRESS	Pass
7	Tracker starts at phase REQUIREMENT_FEASIBILITY	REQUIREMENT_FEASIBILITY	REQUIREMENT_FEASIBILITY	Pass
8	advancePhase() moves the tracker to DEVELOPMENT	DEVELOPMENT	DEVELOPMENT	Pass
9	Status becomes COMPLETED after the full life cycle	COMPLETED	COMPLETED	Pass
10	A data-analytics project bills more than a web project	bill(1.6) > bill(1.2)	bill(1.6) > bill(1.2)	Pass
11	Web project invoice equals 1500 × 1.2	1800.00	1800.00	Pass
12	Feedback rating of 9 is rejected	IllegalArgumentException	IllegalArgumentException	Pass
13	Average of ratings 4 and 2	3.0	3.0	Pass
14	findEmployeesWithSkill("Go") matches one employee	1	1	Pass

No.	Test case	Expected	Observed	Result
15	totalRevenue() over two projects (1000×1.2 + 1000×1.6)	2800.00	2800.00	Pass

Table 2: Test cases and their results.

The output produced by the program is shown in Figure 2. The upper portion is the report of the demonstration scenario and the lower portion is the result of the test suite.

```

Terminal - ProjectManagementDemo

=====
PROJECT MANAGEMENT SYSTEM - DEMO SCENARIO
=====

Company: Acme Software Pvt. Ltd.
Total clients : 2
Total employees: 5
Total projects : 2

=====
WebApplicationProject "RetailKart Storefront" [P-01] client=RetailKart status=COMPLETED type=Web Application
Requirement : Requirement[dept=E-Commerce, feasible=true]: Online storefront with cart & payments [Product catalogue, Shopping cart, Payment gateway]
Life cycle : Tracker[current=Maintenance] Requirement & Feasibility=100% Development=100% Testing=100% Deployment=100% Maintenance=0%
Completion : 80%
Assignments (which employee handles this project):
- Alice handles "RetailKart Storefront" as Project Manager (since 2026-07-08)
- Bob handles "RetailKart Storefront" as Lead Developer (since 2026-07-08)
- Dan handles "RetailKart Storefront" as QA Engineer (since 2026-07-08)
Client feedback at each stage:
- [Requirement & Feasibility] 5/5 - "Scope looks great" (2026-07-08)
- [Development] 4/5 - "Good progress on cart" (2026-07-08)
- [Testing] 4/5 - "Few bugs, fixed quickly" (2026-07-08)
- [Deployment] 5/5 - "Smooth go-live" (2026-07-08)
Avg feedback: 4.50 / 5
Billing[base=5000.00, hours=180.0, rate=55.00, tax=18%, paid=false]
Invoice (incl. tax, complexity x1.2): 21098.40

=====
MobileAppProject "MediCare Patient App" [P-02] client=MediCare Labs status=IN_PROGRESS type=Mobile Application
Requirement : Requirement[dept=Healthcare, feasible=true]: Patient appointment & records app [Appointment booking, Lab report viewer]
Life cycle : Tracker[current=Requirement & Feasibility] Requirement & Feasibility=0% Development=0% Testing=0% Deployment=0% Maintenance=0%
Completion : 0%
Assignments (which employee handles this project):
- Carol handles "MediCare Patient App" as Android Developer (since 2026-07-08)
- Eve handles "MediCare Patient App" as DevOps (since 2026-07-08)
Client feedback at each stage:
- [Requirement & Feasibility] 4/5 - "Clear requirements" (2026-07-08)
Avg feedback: 4.00 / 5
Billing[base=7000.00, hours=90.0, rate=60.00, tax=18%, paid=false]
Invoice (incl. tax, complexity x1.4): 20484.80

=====
TOTAL COMPANY REVENUE: 41583.20
=====

=====
TEST CASES
=====

[PASS] Employee.hasSkill finds an existing skill
[PASS] Employee.hasSkill rejects a missing skill
[PASS] Assignment appears on the project
[PASS] Assignment appears on the employee
[PASS] Project starts NOT_STARTED
[PASS] Project becomes IN_PROGRESS after first assignment
[PASS] Tracker starts at REQUIREMENT_FEASIBILITY
[PASS] Tracker advances to DEVELOPMENT
[PASS] Project is COMPLETED after full life cycle
[PASS] Higher complexity yields higher bill
[PASS] Web invoice equals 1500 * 1.2 = 1800
[PASS] Feedback rating outside 1.5 is rejected
[PASS] Average feedback is (4+2)/2 = 3.0
[PASS] findEmployeesWithSkill returns only matching employees
[PASS] totalRevenue aggregates all projects

=====
RESULT: 15 passed, 0 failed, 15 total
=====

```

Figure 2: Output of ProjectManagementDemo.

7. Graphical User Interface

A desktop front end was written over the same model classes using Swing, which is part of the standard library, so that no external dependency is introduced. The window is divided into a navigation panel on the left and one of four views on the right. The dashboard summarises the portfolio, the project view shows the life cycle, team, requirement, feedback and billing of each project, and the remaining two views list the employees and the clients. The interface can be displayed in a light or a dark appearance.

The project view is not merely a display. Pressing the button marked *Advance life cycle* calls `Project.advanceLifeCycle()` on the underlying object, and the progress bars, the status and the invoice are redrawn from the new state of the model.

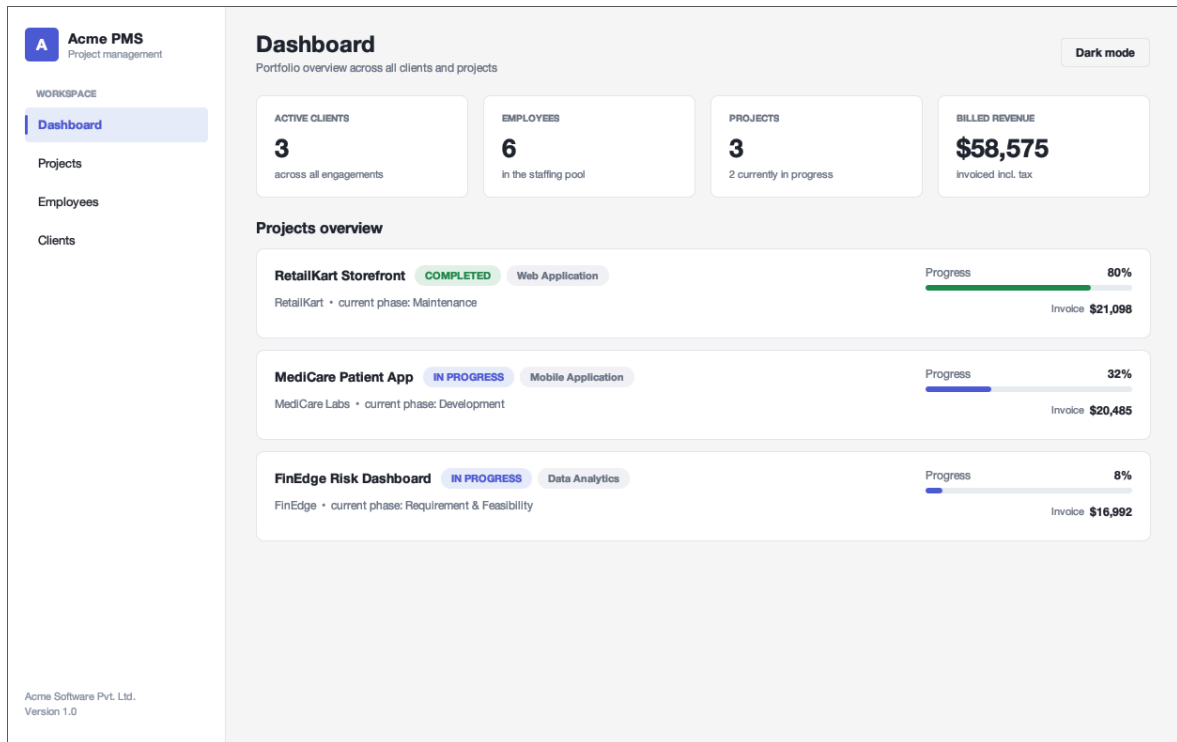


Figure 3: Dashboard view.

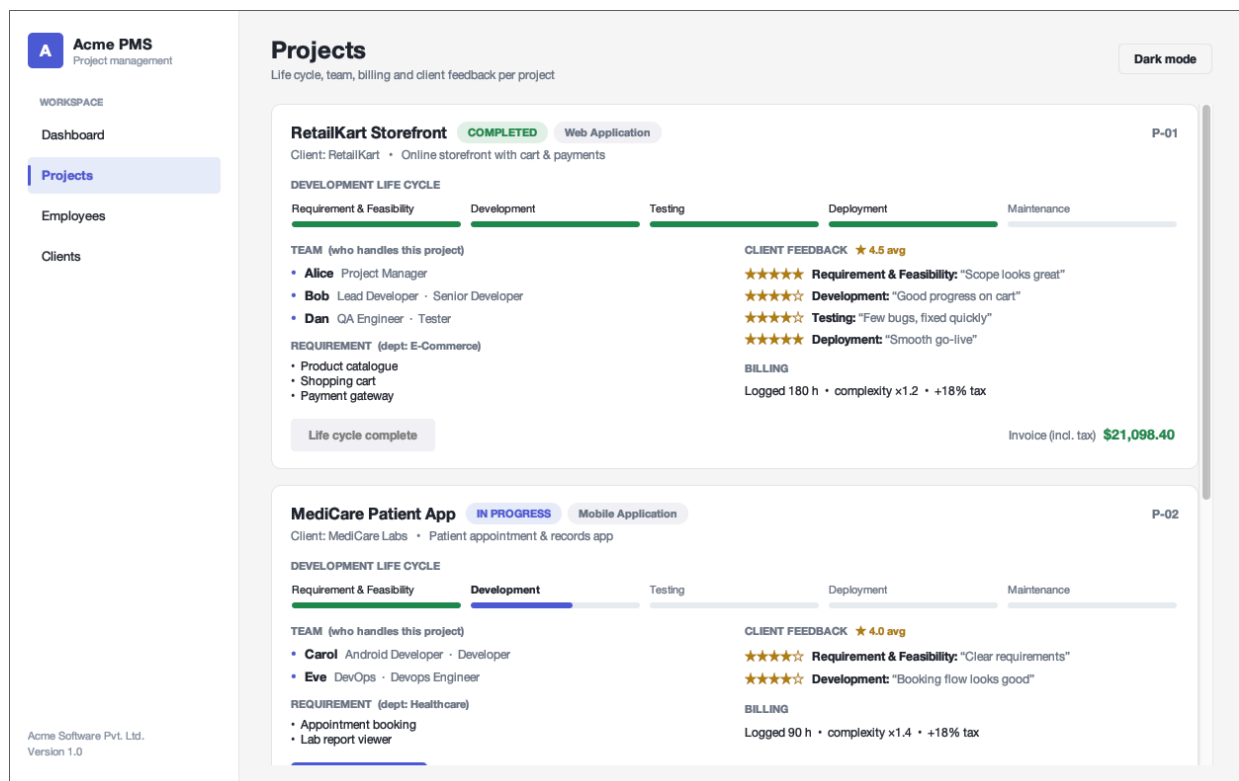


Figure 4: Project view, showing the life-cycle tracker, the assigned team, the requirement, the client feedback and the billing of each project.

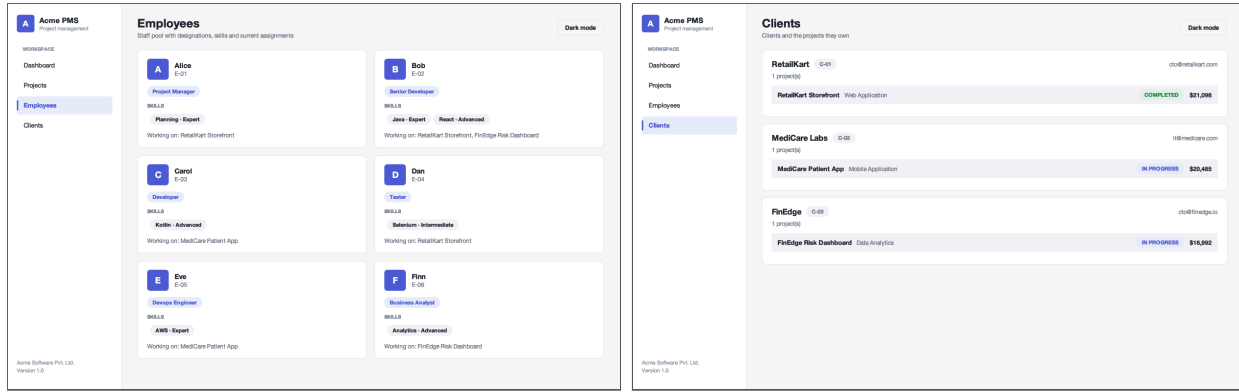


Figure 5: Employee view (left) and client view (right).

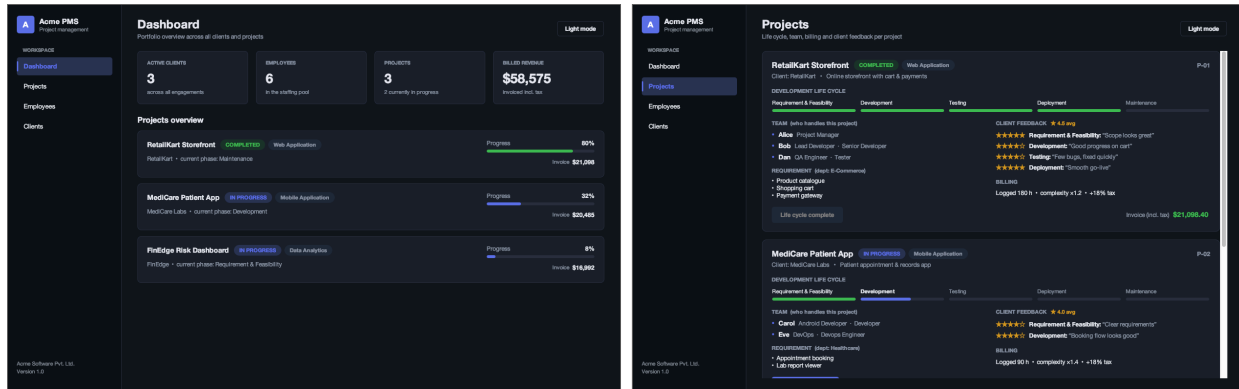


Figure 6: The same two views in the dark appearance.

8. Results and Discussion

All fifteen test cases pass, and the demonstration scenario produces the expected invoice of 21,098.40 for the web project and 20,484.80 for the mobile project. The difference between the two follows from the complexity factors 1.2 and 1.4 respectively, which confirms that the abstract method is being resolved to the correct subclass at run time.

Taking `Project` as an abstract super class keeps the cost of change low. A new type of project requires one small subclass and no change anywhere else, because every other class refers to the super class only. Similarly, a new life-cycle phase or a new designation requires only a new constant in the corresponding enumeration.

The chief limitation of the present implementation is that all data is held in memory and is lost when the program terminates. The billing model is also a simplification, in that it applies a single blended hourly rate to a project rather than charging each employee at the rate of their own designation, although the rate is already stored against the designation and could be used directly.

9. Source Code and Execution

The complete source code, the UML source file, the build scripts and the screenshots reproduced in this report are available in the following public repository:

<https://github.com/Codewithsayanjib/project-management-system-uml>

The project has no external dependencies, so a Java Development Kit (version 11 or later) is the only requirement. To compile and run it:

- Clone the repository and change into it: `git clone <repository URL> && cd project-management-system-uml`
- Compile all the sources into the out directory: `find src -name "*.java" > sources.txt && javac -d out @sources.txt`
- Run the demonstration and the test suite: `java -cp out com.pms.app.ProjectManagementDemo`
- Build and launch the graphical front end: `./build_app.sh` followed by `java -jar dist/AcmePMS.jar`